

Alkanes & Cycloalkanes

1. What is the general formula for an alkane?
2. What is the general formula of a saturated cycloalkane?
3. Why are alkanes saturated?
4. Are all cycloalkanes saturated? Explain.
5. How many hydrogen atoms are in the chemical formula of an alkane that has 3 methyl branches on a 6-carbon chain?
6. How many hydrogen atoms are in the chemical formula that has 2 methyl branches on a 5-carbon chain cyclic main chain?
7. Why are the MP/BP of alkanes & cycloalkanes typically low?
8. Why does the MP/BP of alkanes & cycloalkanes increase with mass?
9. For structural isomers, why do branched alkanes typically have slightly lower MP/BP than non-branched alkanes?

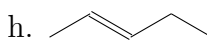
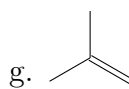
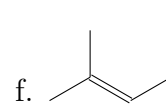
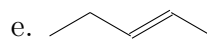
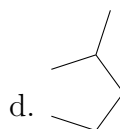
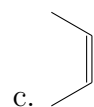
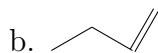
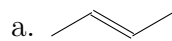
10. Why are alkanes & cycloalkanes soluble in oil but not in water?
11. For structural isomers, why do straight chain alkanes have slightly higher densities than branched alkanes?

Alkynes

1. What do all alkynes have in common?
2. Are alkynes saturated? Explain.
3. How many hydrogens are in a structure that has 3 triple bonds in a 8-carbon chain?
4. Why are alkynes linear?
5. Why do cycloalkynes not really exist?

Geometric Isomers & Alkenes

1. What does it mean for a molecule to have a geometric isomer?
2. Identify whether each of the following structures is a geometric isomer. If yes, draw the other geometric isomer.

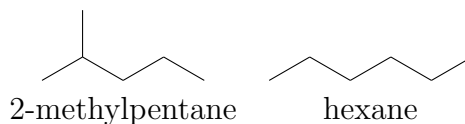


3. What do all alkenes have in common?
4. Are alkenes saturated? Explain.
5. How many hydrogens are in a chemical formula with 2 double bonds on a 5-carbon chain?
6. How many hydrogens are in a chemical formula with 2 cyclics, 1 triple bond & 3 double bonds on a 20-carbon chain?

General Physical Properties of Aliphatics

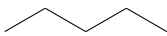
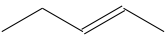
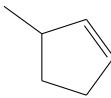
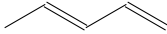
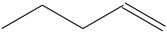
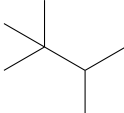
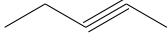
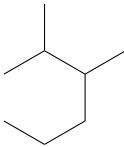
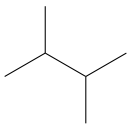
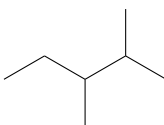
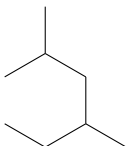
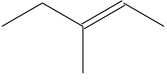
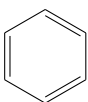
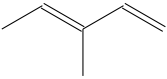
1. What does it mean for two molecules to be structural isomers?
2. What happens to two molecules during a change of state?
3. There are three structural isomers of C_5H_{12} . Their boiling points vary, with values of $36.1^\circ C$, $27.8^\circ C$ & $9.5^\circ C$. Draw each isomer, indicate its expected boiling point & explain its boiling point relative to the other isomers.
4. How would the densities of these three isomers of C_5H_{12} vary? Explain your answer.
5. All hydrocarbons undergo complete combustion reactions to produce $CO_2(g)$, $H_2O(l)$ & energy. Would the number of CO_2 & H_2O molecules be expected to be the same from burning one gram of each isomer of C_5H_{12} ? Explain your answer.
6. Would the amount of heat produced be the same when one gram of each isomer of C_5H_{12} undergoes combustion? Explain.
7. Small aliphatic compounds tend to be found in the gas state at room temperature. Explain.

8. When comparing 2-methylpentane with hexane:
- Which should have the higher boiling point & why?
 - Which should have the higher density & why?



Aliphatic Compound Review

Identify each functional group in each molecule. (Alkane is the default if no other functional group is present.)

- | | | |
|--|---|---|
| 1.  | 8.  | 16.  |
| 2.  | 9.  | 17.  |
| 3.  | 10.  | 18.  |
| 4.  | 11.  | 19.  |
| 5.  | 12.  | 20.  |
| 6.  | 13.  | |
| 7.  | 14.  | |
| | 15.  | |

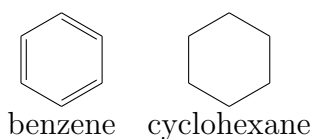
Identify each functional group in each molecule.

- | | |
|------------------------------------|----------------------------------|
| a. 2,3-dimethylpentane | i. 5-butyl-2,4,7-trimethyldecane |
| b. 2-methylbutane | j. 3,5-diethyl-1-octyne |
| c. 5-ethyl-1,3-cyclopentadiene | k. 4,5,6-trimethylnonane |
| d. 3,4-diethyl-2,5-dimethyloctane | l. 2,3,4,4-tetramethyl-2-hexene |
| e. 2,3,4-trimethyl-trans-3-hexene | m. 1,3,5-cyclohexatriene |
| f. 3,4-diethyl-1-trans-5-octadiene | n. 3-methyl-1,2-butadiene |
| g. 4-ethyl-3-methyl-cis-2-heptene | o. 3-methyl-trans-2-hexene |
| h. 1,3-dimethylcyclopentane | p. 2-methyl-1-propene |

- q. 3-ethyl-1-pentyne
- r. cis-2-pentene
- s. 1,3-cyclooctadiene
- t. 3-methyl-cis-2-pentene
- u. 3,4-dimethyl-1-cyclooctene
- v. 1-trans-3,5-hexatriene
- w. 4-ethyl-3-propyl-1-hexyne
- x. 1,2,3,4-tetramethylcyclobutane
- y. cis-2-trans-4-hexadiene

Benzene-Based (Aromatic) Hydrocarbons

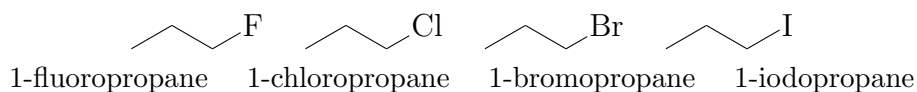
1. What do all aromatic hydrocarbons have in common?
2. When comparing benzene to cyclohexane:
 - a. Which should have the higher boiling point & why?
 - b. Which should have the higher density & why?



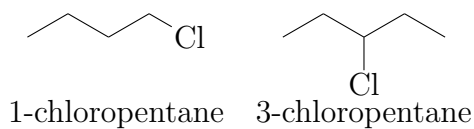
3. Why is the MP/BP of benzene typically low?
4. Why is benzene soluble in oil but not in water?
5. Why do aromatics have a higher density than alkanes with the same number of carbons?

Alkyl Halides

1. What do all alkyl halides have in common?
2. Which of 1-fluoropropane, 1-chloropropane, 1-bromopropane or 1-iodopropane should have the higher boiling point & why?



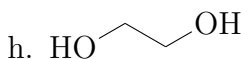
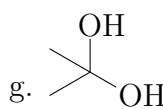
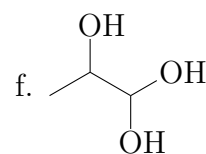
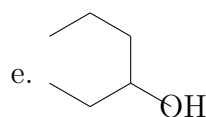
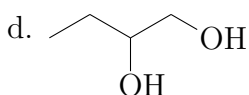
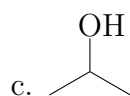
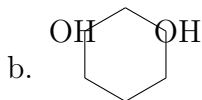
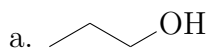
3. Without looking up the boiling point data for 1-chloropentane & 3-chloropentane, provide a valid argument supporting why:
- 1-chloropentane should have the higher boiling point
 - 3-chloropentane should have the higher boiling point



4. Why do alkyl halides have higher MP/BP than their corresponding aliphatics (& aromatics)?
5. Why are alkyl halides more soluble in water than aliphatics?
6. Why does adding an alkyl halide to water cause the solution to freeze at lower temperatures?

Alcohols

1. What do all alcohols have in common?
2. Distinguish each OH as 1°, 2° or 3°.



3. (Challenging): Distinguish each OH as 1°, 2° or 3°.

- | | |
|------------------------------------|----------------------------------|
| a. 2-methyl-2-propanol | e. 3-cyclopropyl-2-hydroxyhexane |
| b. 1,4-dicyclobutyl-2,3-butanediol | f. 2-hexanol |
| c. phenylmethanol | g. 3-hydroxypentane |
| d. 1,2-ethanediol | h. 2-hydroxyhexane |

4. How do you know which OH is 1°, 2° or 3°?

5. Assuming both have the same number of carbons, would an alcohol with more OH groups be more soluble in water than alcohols with only one OH group?

6. Explain the MP/BP trend: aliphatics < alkyl halides < alcohols.
7. Why does exposure increase MP/BP?
8. Why are alcohols generally soluble in water?
9. Why is a 1-carbon alcohol more soluble than a 10-carbon alcohol in water?
10. Why does adding an alcohol to water cause the mixture to boil at a higher temperature?

Ethers

1. What do all ethers have in common?
2. Which of alcohols or ethers should have the higher boiling point & why?
3. Explain the MP/BP trend: aliphatics < ethers < alcohols.
4. Why are ethers good solvents of polar, non-polar & ionic solutes?
5. Why does the solubility of a large ether lower than the solubility of a smaller ether?

Aldehydes

1. What do all aldehydes have in common?
2. Is it necessary to write the number “1” for the location of the carbonyl group in aldehydes? Explain.
3. Unlike alkanes, explain why the boiling points of aldehydes do not increase drastically as the number of carbons increase.
4. Which of 1° alcohols or aldehydes should have the higher boiling point & why?
5. Explain the MP/BP trend: aldehydes < alcohols.
6. Why are aldehydes less water soluble than alcohols?
7. Are aldehydes good general solvents of ionic, polar & non-polar solutes? Explain.

Ketones

1. What do all ketones have in common?
2. Under what circumstances is it possible to have the name “ethanone” as part of a molecule name?

3. Is it necessary to include the location number of the carbonyl group in ketones? Explain.
4. Explain the MP/BP trend: for large structural isomers, aldehydes $\doteq$ ketones $<$ alcohols.
5. Why are ketones less soluble in water than alcohols?
6. Are ketones good general solvents of ionic, polar & non-polar solutes? Explain.

Carboxylic Acids

1. What do all carboxylic acids have in common?
2. Is it necessary to write the location numbers for the location of the carboxyl group(s) in carboxylic acids? Explain.
3. Unlike aliphatics, small carboxylic acids tend to be found in the solid state at room temperature. Explain.
4. Explain the MP/BP trend: alcohols $<$ carboxylic acids.
5. Why does the MP/BP of carboxylic acids increase with the number of carboxyl groups?

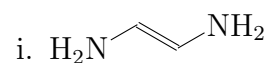
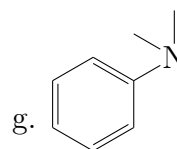
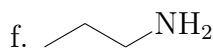
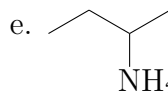
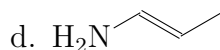
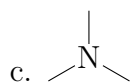
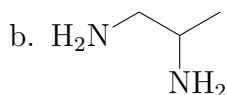
6. Why are smaller carboxylic acids more soluble than larger carboxylic acids?
7. Why do carboxylic acids tend to be solid state than gas state?

Esters

1. What do all esters have in common?
2. Would small esters be found in the gas or solid state at room temperature? Explain.
3. Would small esters be soluble in water? Explain.
4. Explain the MP/BP trend: esters < carboxylic acids.
5. Why are most esters generally insoluble in water?
6. Why are esters less polar than carboxylic acids?

Amines

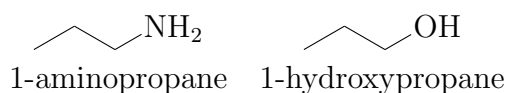
1. What do all amines have in common?
2. Identify each N as 1°, 2° or 3°:



3. (Challenging) Identify each N as 1°, 2° or 3°:

- | | |
|-------------------------------------|-----------------------|
| a. N-cyclopentyl-N-methylhexanamine | d. 2,2-diaminohexane |
| b. 1,2-diaminoethane | e. benzenamine |
| c. 2-aminobutane | f. N-methylbutanamine |

- Distinguish between a 1°, 2° & 3° amine.
- Assuming they are structural isomers, which of 1°, 2° or 3° amines be the most soluble in water? Explain.
- Assuming they are structural isomers, which of 1°, 2° or 3° amines be the most soluble in benzene? Explain.
- Which of 1-aminopropane or 1-hydroxypropane should have the higher boiling point & why?



- Explain why amines have higher MP/BP than aliphatics.
- For amine structural isomers, why do 1° > 2° > 3° for MP/BP?
- Why are small amines soluble in water?

Amides

1. What do all amides have in common?
2. Explain why amides are rarely gases at room temperature.
3. Why are smaller amides more soluble than larger amides?
4. Why do amides tend to be solids or liquids at room temperature?

Organic Chemistry Semi-Unit Review (Part 1) - Structures & Properties

The following is a small selection of questions that, if used appropriately, can better help you prepare for the unit test. This is not a complete review sheet. Answer the following questions on a separate piece of paper, using correct format.

Multiple Choice:

1. The functional group of an amide is:
 - a. -OH
 - b. -C=C
 - c. -NR_2
 - d. -CONR_2
 - e. -OR
2. Carbon atoms can bond with each other to form:
 - a. single bonds only
 - b. double bonds only
 - c. single & double bonds
 - d. single, double & triple bonds
 - e. single, double, triple & quadruple bonds

Short Answer & Full Solution:

1. Identify the functional group(s) for each. (Alkane is the default if no other functional groups exist on the structure).

